MIL-C-44172(GL)
30 June 1985

MILITARY SPECIFICATION

CAULIFLOWER AU GRATIN, POWDER
(FOR ORAL LIQUID FEEDING)

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers cauliflower au gratin powder for oral liquid feeding. This product is one component of a meal that, when consumed with other components in three meals or supplementary nourishments, provides a nutritionally adequate, orally consumed diet.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-P-44151	- Packaging and Meal Assembly of Powder (For Oral Liquid Feeding)

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Frozen Cauliflower

(Application for copies should be addressed to Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Cheddar Cheese

U.S. Standards for Grades of Nonfat Dry Milk (Spray Process)

Dairy Plants Surveyed and Approved for USDA Grading Service

(Application for copies should be addressed to Dairy Standardization Section, Dairy Division, Room 2750-So, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

NATIONAL CHEESE INSTITUTE

Cheese Color Standards

(Application for copies should be addressed to the National Cheese Institute, 699 Prince Street, Alexandria, VA 22314.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP) and the National Formulary (NF)

(Application for copies should be addressed to the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Cauliflower, frozen. Frozen cauliflower shall be U.S. Grade A as specified in the U.S. Standards for Grades of Frozen Cauliflower.

3.2.2 Margarine. The margarine shall conform to the Federal Food, Drug, and Cosmetic Act Standards of Identity, 21 CFR 166.40 and 166.110. The margarine shall have a sweet pleasing taste and aroma, free from rancidity or other objectionable flavors and odors. It shall have a medium yellow color and a smooth body and texture free from crumbliness, graininess, grittiness, oiliness, or droplets of free moisture. The margarine shall be salted. The margarine shall be in excellent condition, free from mold, slime, or discoloration. The margarine shall be manufactured in a plant listed in the Dairy Plants Surveyed and Approved for USDA Grading Service.

3.2.3 Cheese, cheddar. The cheese shall be mild cured U.S. Grade A or better cheddar cheese as defined in the U.S. Standards for Grades of Cheddar Cheese. The cheddar cheese shall be grated and medium yellow-orange in color (No. 5 to No. 8 when compared to the National Cheese Institute's Cheese Color Standard). The manufacturer shall grate the cheese.

3.2.4 Milk, nonfat dry (low heat). Nonfat dry milk shall be U.S. Extra Grade, Low Heat as defined in the U.S. Standards for Grades of Nonfat Dry Milk (Spray Process).

3.2.5 Salt. Salt shall be iodized, clear white, refined chloride with or without anticaking agents.

3.2.6 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L., from which the outer covering or the outer and inner coverings have been removed. The ground white pepper shall have a characteristic odor; a hot, biting, pungent flavor; and a light color.

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The white pepper shall contain not less than 0.1 mL volatile oil per 100 grams and shall be of such a size that not less than 95 percent passes through a U.S. Standard No. 40 sieve.

3.2.7 Starch, freeze-thaw stable. Starch shall be modified waxy maize of the type used in frozen processed foods. The cooked starch slurry shall be bland, with essentially no cereal or starchy taste. It shall be white, odorless, and finely pulverized.

3.2.8 Methylcellulose. Methylcellulose shall be a white, free-flowing powder with no off odors or flavors. It shall have a viscosity of 4,000 centipoises in a 2 percent aqueous solution as measured by a temperature of 68°F Ubbelohde tube for viscosity measurements. The viscosity shall be certified by a certificate of conformance provided by the manufacturer. The methylcellulose shall comply with the requirement of the Food Chemicals Codex.

3.2.9 Water. Water for formulation and washing shall comply with the National Interim Primary Drinking Water Regulations.

3.3 Preparation and processing.

3.3.1 Product formulation. The formula for cauliflower au gratin powder for oral liquid feeding shall be as follows:

Ingredients	Percent by weight
Water	42.89
Cauliflower, frozen	39.00
Margarine	2.00
Cheese, cheddar	8.70
Milk, nonfat dry	2.00
Salt	0.20
Pepper, white, ground	0.01
Starch, freeze-thaw stable	0.50
Water, cold, for starch slurry	0.50
Methylcellulose	0.20
Water, hot (170°-194°F), for methylcellulose	2.00
Water, cold (40°-50°F), for methylcellulose	2.00

3.3.1.1 Amount of water. The proportion of water in the formulation is specified to enable pumping of the product. This amount may be increased or decreased to facilitate pumping or to change the viscosity of the product. Changes in the amount of water should be within limits necessary to dissolve th ingredients without unnecessarily increasing the amount of water that must be removed in the freeze dehydration process.

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3.3.2 Cooking. The cauliflower shall be cooked in water until very soft and tender, then drained, reserving the cooking water. The cauliflower shall be pureed or ground through a plate having holes of 1/16-inch diameter. The margarine shall be melted and to it shall be added the reserved cooking water, grated cheese, nonfat dry milk, salt, and pepper. The mixture shall be heated slowly while being stirred, until the cheese melts and the temperature reaches 180°F. A slurry shall be made with the starch and a small amount of cold water; this slurry shall be added to the cheese mixture and the temperature brought back to 180°F. The ground cauliflower shall be added. The mixture shall be heated to 180°F and held at that temperature for 10 minutes. Then the heat shall be turned off and the mixture cooled rapidly to 120°F. Methylcellulose shall be prepared for addition by mixing with hot water (176°-194°F) until all particles are wetted and a smooth paste is obtained. The cold water (40°-50°F) shall be added to this methylcellulose paste, and stirring shall continue until a smooth slurry is obtained. The methylcellulose slurry shall be combined with the other ingredients and stirred until the mixture is smooth. The mixture shall be frozen to 0°F or below within 12 hours after cooking is completed.

3.3.3 Freezing. The product shall be portioned onto freeze dryer trays immediately after cooling and shall be frozen to 0°F or below within 12 hours after cooking is completed. The product shall be freeze dehydrated within 48 hours after being frozen.

3.3.4 Freeze dehydration. The product shall be freeze dehydrated at an absolute pressure not to exceed 1.5 millimeters of mercury. The product temperature, as indicated by a suitable instrument, shall be no more than 130°F, and the platen temperature shall be 135°F. After dehydration has been completed, the pressure shall be equalized to atmospheric level with nitrogen, and the product shall be pulverized. During the interim, the product shall be adequately protected from oxygen and moisture by being held either under a nitrogen atmosphere of 2.0 percent oxygen or less or under a vacuum of at least 27.0 inches of mercury for the entire period. If a vacuum is used, it shall be broken with nitrogen. The nitrogen shall comply with U.S. Pharmacopeia grade as specified in the United States Pharmacopeia.

3.3.5 Pulverization. After freeze dehydration, the product shall be pulverized through a U.S. Standard No. 10 screen (0.0787 inch).

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3.3.6 Packaging. After pulverization, the product shall be packaged without delay in pouches as specified in MIL-P-44151. In no case shall more than 16 hours elapse from the time the freeze dehydration chamber is opened to the time the product is completely packaged, and at no time shall more than 3 hours elapse from the time the product is pulverized to the time it is completely packaged.

3.4 Finished product requirements.

3.4.1 Dehydrated, pulverized product. The dehydrated product shall comply with the following requirements:

a. The product shall contain no foreign or extraneous material.

b. The product shall be free from caking and hard lumps.

c. The product shall leave no particles remaining when passed through a U.S. Standard No. 10 sieve.

d. The moisture content shall not exceed 2.0 percent (see 4.5.1).

e. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample (see 4.5.2).

f. The Escherichia coli count shall be less than 3 per gram (no positives in the standard 3 tube MPN technique) in four of five samples and not greater than 20 per gram in any sample (see 4.5.2).

g. The sample average net weight of the lot shall be not less than 30.0 grams of product.

h. The invidual pouches shall contain not more than 33.0 grams nor less than 27.0 grams.

3.4.2 Rehydrated product. Rehydration of the dehydrated powder for oral liquid feeding shall be accomplished by mixing each pouch (30 grams) with 1 cup (8 ounces) of hot water in a high-speed food blender for 30 seconds. For a pureed diet, only 1/2 cup (4 ounces) of liquid should be used. The rehydrated product shall comply with the following requirements:

a. After rehydration for oral liquid feeding, no more than 0.5 gram of product shall remain when the product has passed through a U.S. Standard No. 40 sieve.

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4.3. First article inspection. When a first article is required (see 6.2), it shall be inspected in accordance with the quality assurance provisions of MIL-P-44151 and this document and shall be evaluated for overall appearance and palatability. Any failure to conform to the required quality assurance provisions or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall comply with MIL-STD-105. The finished product shall comply with the quality assurance provisions of MIL-P-44151 and this document.

4.4.1 Component and material examination. In accordance with 4.1, examination of components and materials shall comply with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents.

4.4.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance shall be evident by examination of pertinent labels, markings, U.S. Grade certificates, certificates of analyses, or other such supplier-furnished valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically as necessary, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.4.1.2 Filled and sealed pouch examination. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44151.

4.4.2 In-process examination. In-process examination shall be performed to determine conformance to preparation and processing requirements in 3.3. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.4.3 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample on a suitable scale tared with a representative empty pouch. Results shall be reported to the nearest 0.1 gram. Any individual net weight of less than 27.0 grams or more than 33.0 grams shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The

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inspection level shall be S-3, and the acceptable quality level (AQL) shall be 2.5, expressed in terms of percent defective. In addition, the lot shall be rejected if the sample average net weight is less than 30.0 grams.

4.4.4 Finished product examination. The finished product shall be inspected for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3, and the AQL shall be 1.5, expressed in terms of defects per hundred units.

TABLE I. Product characteristics 1/ 2/

Category	Defect
Major	Applicable to dehydrated product
101	Product is not free from caking and hard lumps.
102	Presence of particles remain on a U.S. Standard No. 10 sieve. 3/
	Applicable to rehydrated products 4/
103	More than 0.5 gram of product does not pass through a U.S. Standard No. 40 sieve. 5/
104	The rehydrated product does not have flavor or odor normal for cauliflower au gratin.
105	The rehydrated product does not have an off-white color.
106	The viscosity is not acceptable for appropriate use of the product.

1/ The presence of any foreign material (e.g., glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale, etc.), or foreign color shall be cause for the rejection of the whole lot.

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TABLE I. Product characteristics - Continued

2/ Product not equal to or better than the approved preproduction sample (see 6.2) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by a USDA inspection agent.)
3/ The powder shall be poured onto a U.S. Standard No. 10 sieve, and compressed air shall be blown through the product. The presence of visible particles remaining on the sieve surface shall be considered a defect.
4/ Rehydration of the dehydrated powder for oral liquid feeding shall be accomplished by mixing each pouch (30 grams) with 1 cup (8 ounces) of hot water in a high-speed blender for 30 seconds. For a pureed diet, use 1/2 cup (4 ounces) of liquid.
5/ Particle size determination shall be made on the sample(s) after rehydration for oral liquid feeding (use a different sample for organoleptic examination). The rehydrated product shall be poured onto a U.S. Standard No. 40 screen. The product shall be washed for 1 1/2 minutes with water at a temperature of 96° to 100°F that is running at a flow of 90 to 100 mL/sec through 1/2-inch tubing (interior dimension). The sieve shall be allowed to drain for two minutes, then excess water may be wiped from bottom and sides of the sieve. The remaining solids shall be blown from the sieve with compressed air onto a tared weighing paper (see 3.4.2).

4.4.5 Packaging. Packaging shall be in accordance with the applicable requirements of MIL-P-44151.

4.5 Product testing. The dehydrated powder shall be tested for moisture content as specified in 4.5.1 and for microbiological requirements as specified in 4.5.2.

4.5.1 Moisture content testing. The product shall be tested for moisture content in accordance with the Moisture in Dried Fruits Method (vacuum drying) in the Fruit and Fruit Product Chapter in the Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC) with the exception that the product shall be dried for 16 hours. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4.1

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shall be considered a defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed laminated pouch. The inspection level shall be S-2 and the AQL shall be 1.5, expressed in terms of percent defective.

4.5.2 Microbiological testing. Five filled and sealed pouches of the product shall be selected at random from the lot regardless of lot size. The contents of each sample pouch shall be microbiologically tested, as applicable, in accordance with the Microbiological Methods Chapter of the Official Methods of Analysis of the AOAC using the section on Examination of Frozen, Chilled, Precooked, or Prepared Foods-Official First Action. The diluent shall be added to each sample of dry product and allowed to stand for 15 minutes before the blending of that sample. Continue as directed by AOAC. Any test failure shall be cause for rejection of the lot. Any of the following results shall be considered a test failure:

a. Any of the five samples has an aerobic plate count greater than 150,000 per gram.

b. Two of the five samples have an aerobic plate count greater than 75,000 per gram.

c. Any of the five samples has a confirmed E. coli count of greater than 20 per gram.

d. Two of the five samples have a confirmed E. coli count equal to or greater than 3 per gram.

5. PACKAGING

5.1 Packaging. Packaging shall comply with the applicable requirements of MIL-P-44151.

5.2 Labeling and marking.

5.2.1 Pouches. The pouches shall be labeled as follows and examined for compliance in accordance with MIL-P-44151.

a. CAULIFLOWER AU GRATIN, POWDER (FOR ORAL LIQUID FEEDING)

b. MEAL NO. 5 (LUNCH)

c. List of ingredients in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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d. Mixing instructions, as follows:
"Empty contents of pouch into a high-speed food blender.
Add 1 cup (8 ounces) hot water.
Blend for 30 seconds.
(For a pureed diet, use 1/2 cup (4 ounces) of liquid.)"

e. The net weight, as follows: 30 grams net."

f. Contractor's name and address.

g. USDA inspection marking (as required).

h. Date markings in accordance with MIL-L-1497.

i. Code.

6. NOTES

6.1 Intended use. This product is intended to be used by patients requiring an oral liquid or pureed diet (e.g., those with maxillofacial injuries, wired jaws, oral surgery, dental disease, or swallowing difficulties).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Provisions for preproduction samples (see 3.4.3 and 6.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Custodian:

Army - GL

Review activities:

Army - MD, TS
DP - SS

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44172(GL)	Cauliflower au gratin, Powder (For Oral Liquid Feeding)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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